

parameter estimation

computing and differentiating efficiently

Motivation: OT as a Loss Function

- Statistical divergences often used as a loss in **parameter estimation** problems
- OT has many 'nice' properties, motivating its use as a loss function
- Main challenge: **computing and differentiating efficiently**
- Various applications: synthesis, image segmentation, inverse problems, and shape matching
- Particularly effective when data lies on low-dimensional manifolds or lacks a well-defined density

OT as a *Variational* Objective